Year 8

Mini Test 1

Number

Total Marks: 33 marks
Total Time: 30 minutes

Name: ______________________

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks.

Calculator Not Allowed

Question 1	**4 marks**

Match the correct definition to each of the following terms:

a	Integer		A positive number that multiplies by another positive number to give a product.
b	Multiple		Has exactly 2 factors. 1 and itself.
c	Factor		The result of multiplying a number by an integer.
d	Prime number		A positive or negative whole number.

(1 mark each)

Question 2	4 marks

Number table:

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20

Use the number table to answer the following questions:

a) List four PRIME numbers: (2 marks)

b) List two SQUARE numbers: (1 mark)

c) List all the multiples of 4: (1 marks)

Question 3	6 marks

Use index laws to ***simplify*** *the following:*

a) $2 \times 2 \times 2 \times 2 \times 3 \times 3 \times 3 =$ (1 mark)

b) $(5^2)^3$ (1 mark)

c) $3^2 \times 2^3 \times 7 \times 3$ (1 mark)

d) $3^7 \div 3^2 \div 3$ (1 mark)

e) $6 \times 2^4 \div (3 \times 2^2) \times 5$ (2 marks)

Question 4	**6 marks**

***Evaluate** the following:*

a) 4^3 (1 mark)

b) $\sqrt{81}$ (1mark)

c) 100^2 (1 mark)

d) $\sqrt[3]{64}$ (1 mark)

e) $2^3 \times 2^2$ (1 mark)

f) 7^0 (1 mark)

Question 5	4 marks

Use a factor tree to solve the following question ***and then use index notation*** *to write 360 as a product of its prime factors:*

360

Question 6	3 marks

Find the highest common factor (HCF) of the following:

a) 24 and 36 (2 marks)

b) 7 and 13 (1 mark)

Question 7	**4 marks**

Find the lowest common multiple (LCM) of the following:

a) 7 and 3 (2 marks)

b) 8 and 12 (2 marks)

Question 8	**2 marks**

On Monday, Billy borrows \$40 from his friend. On Tuesday, he pays his friend \$35. On Friday morning, he borrows \$47 from Billy and pays him back \$39 later that night. How much does Billy owe his friend?

END OF TEST